Dr. Ingo Frommholz

Professor of Applied Data Science
PhD (Dr. rer. nat.)
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At a Glance

- Full Professor and Head of School of Applied Data Science, Modul University Vienna, Austria; Adjunct Professor (Lehrbeauftragter) for Data Engineering at Bern University of Applied Sciences, Switzerland
- Interested in user-oriented Information Retrieval, AI, NLP and Data Engineering with application for example in academic search and recommendation, digital libraries, social sciences, e-commerce, and digital humanities
- Chair of the BCS Information Retrieval Specialist Group (BCS-IRSG)
- Founding Head of the Data Science and AI Research group (DAIREL), University of Wolverhampton, UK (2023–2025)
- Steering committee member, ACM CIKM (Core-A-ranked conference) and ACM SIGIR ICTIR conference series
- Funding (PI): FFG project Trustlabel.AI on hybrid AI for scientific claim verification (2027–2030, total EUR 822,938, own part EUR 245,180); EU Horizon Europe MSCA Staff Exchange project OMINO on Information Overload (2023–2026, UKRI guarantee funding, total EUR 1,550,200, own part EUR 253,000); EU H2020 ETN QUARTZ on quantum mechanics and information retrieval (2016–2020, total EUR 3,461,495, own part EUR 546,756); internally funded RIGHT project on deep learning for digital humanities (2023–2025, GBP 612,621); plus further smaller research and knowledge transfer projects.
- Published more than 100 refereed and unrefereed articles, preprints, reports, proceedings, books and book sections
- Senior Managing Editor of the International Journal on Digital Libraries (Springer)
- General Chair, ACM CIKM 2023, Birmingham (Core-A-ranked conference with 750 delegates) and ELLIS ESSIR 2025; Programme Chair, ACM/IEEE JCDL 2025; Short Paper Programme Chair, ECIR 2027 (Southampton, UK); Organiser of several international workshops and schools (e.g. BIR, BIRNDL, BIRDS, BCS Search Solutions, FDIA, ASIRF)
- Chair and senior programme committee member for several world-leading conferences and journals in Information Retrieval, AI and Digital Libraries
- Course and programme co-ordinator for Computer Science, Data Science and Artificial Intelligence undergraduate and postgraduate programmes; module co-ordinator and lecturer in several Computer Science topics at undergraduate and postgraduate level in the UK, Austria, Germany, Switzerland, Oman and Jordan, such as Information Retrieval, Machine Learning and Artificial Intelligence, Data Mining, Data Science, Cybermetrics and Statistics, Data Engineering, Databases and Data Modelling, Programming and Data Structures, Distributed Computing

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- Supervised 12 PhD students and 2 MSc-by-research students to successful completion; currently supervising 5 PhD students (1 as first supervisor, 4 as external supervisor); examiner for 15 PhD candidates in the UK, Denmark, France, Finland, The Netherlands, Italy, Germany and Austria
- External examiner: University of Sheffield (since 2023); University of Strathclyde (2020–2024)
- PhD in Computer Science (Dr. rer. nat., summa cum laude), University of Duisburg-Essen, Germany
- Fellow of the BCS (FBCS) and Advance HE (FHEA); member of ACM and ACM SIGIR

Personal Information

Main Positions	Full Professor of Applied Data Science, Modul University, Vienna, Austria Adjunct Professor (Lehrbeauftragter) for Data Engineering, Bern University of Applied Sciences, Switzerland
Nationality	German, British
Languages	German (native), English (near-native)

Main Research Interests

- User-oriented and interactive Information Retrieval and Access
- NLP, generative and hybrid AI for Information Retrieval, including Retrieval-Augmented Generation
- Quantum-inspired and geometric models of Information Retrieval
- Bibliometric-enhanced and scholarly Information Access; Digital Libraries
- Database and Information Retrieval Integration; Data and Knowledge Engineering
- Context- and culture-aware Information Retrieval
- Applied Information Retrieval and Data Science, e.g. for the Social Sciences and Digital Humanities

Education

2008 **PhD (Dr. rer. nat.) in Computer Science**, *University of Duisburg-Essen*, Duisburg, Germany
Thesis: *A Probabilistic Framework for Information Modelling and Retrieval Based on User Annotations on Digital Objects*. First examiner: Prof. Dr. Norbert Fuhr, 2nd examiner: Prof. Dr. Maristella Agosti. **German highest grade (summa cum laude)**.

2001 **Diplom-Informatiker (MSc Computer Science equivalent)**, *University of Dortmund*, Germany
Thesis: *Automatic categorisation of Web documents*. Examiner: Prof. Dr. Norbert Fuhr.
Major topics during studies: Operating Systems, Computer Networks & Distributed Systems, Databases, Information Retrieval, Business Informatics. **German highest grade (1.0)**.

2012 **Postgraduate Certificate in Academic Practice (PgCAP)**, *University of Bedfordshire*, Luton, UK
60 credit Masters-level programme designed to both support and enhance the engagement of academic staff with their academic responsibilities.

2007 **Certificate in professional teaching competence in higher education**, *University of Duisburg-Essen*, Duisburg, Germany
The 200 hours training programme addressed teaching and learning, testing, advising and counselling, evaluating, and innovatively developing study programmes and teaching strategies.

Professional Experience

03/2025–Present **Professor of Applied Data Science and Head of School**, *School of Applied Data Science, Modul University*, Vienna, Austria

02/2023–Present **Adjunct Professor (Lehrbeauftragter) for Data Engineering**, *School of Engineering and Computer Science, Bern University of Applied Sciences*, Biel/Bienne, Switzerland

11/2025–Present **Chair (Obmann)**, *Vienna Network for Research and Innovation (VNRI)*, Vienna, Austria

11/2024–Present **Chair**, *BCS Information Retrieval Specialist Group*, London, UK

08/2026–09/2026 **Visiting researcher**, *College of Computing and Data Science, Nanyang Technological University*, Singapore
OMINO Secondment

07/2024–08/2024 **Visiting researcher**, *Department of Computer Science, University of New Hampshire*, Durham, NH, USA
OMINO Secondment

2023–2026 **Visiting researcher**, *GESIS – Leibniz Institute for the Social Sciences*, Cologne, Germany
Several OMINO Secondments

2023–2026 **Visiting researcher**, *OMINO Project Partners*
Several OMINO Secondments at project partners incl. University of Southampton (UK), University of Wolverhampton (UK), University of Strathclyde (UK), Modul University Vienna (Austria), Warsaw University of Technology (Poland), Zeta Alpha Vector B.V. (The Netherlands), Slovenian Press Agency (STA, Slovenia)

12/2020–02/2025 **Reader (Associate Professor equivalent) in Data Science**, *School of Engineering, Computing and Mathematical Sciences, University of Wolverhampton*, UK

02/2011–11/2020 **Lecturer/Senior Lecturer**, *School of Computer Science and Technology, University of Bedfordshire*, Luton, UK
02/2011–07/2012 Lecturer, appointed Senior Lecturer 08/2012

2012–2018 **Search and Data Science Consultant**, *Self Employed*, London, UK

11/2008–01/2011 **Postdoctoral Research Associate**, *Information Retrieval Group, School of Computing Science, University of Glasgow*, Glasgow, UK

10/2004–10/2008 **Research Assistant**, *Information Systems Group, Faculty of Engineering Sciences, University of Duisburg-Essen*, Duisburg, Germany

03/2006–04/2006 **Visiting researcher**, *Information Management Systems Research Group, University of Padua*, Italy

03/2001–09/2004 **Research Assistant**, *Fraunhofer Integrated Publication and Information Systems Institute (IPSI)*, Darmstadt, Germany

11/1999–02/2001 **Software Developer**, *adesso GmbH*, Dortmund, Germany

05/1998–10/1999 **Software Developer**, *o.tel.o GmbH*, Düsseldorf, Germany

08/1994–04/1998 **Software Developer**, *Lion Ges. für Systementwicklung mbH*, Bochum, Germany

Grants, Funding and Projects

Research Grants (as PI / Lead)

Grants to which I contributed significantly in a leading role to the successful bid and/or have been the Principal Investigator (PI) and work package leader.

2027–2030 **TrustLabel.AI**, *Modul University Vienna*, Austria **EUR 822,938**
FFG-funded project (MU part EUR 245,180). Topic: Hybrid AI and retrieval-augmented generation for scientific claim verification in social sciences and climate change communication. Joint project with webLyzard Technology GmbH (Austria), Academy for Climate Action (Austria) and AUSSDA – The Austrian Social Science Data Archive at the University of Vienna (Austria). PI for Modul University and leading proposal author.

2023–2026 **OMINO**, *Univ. of Wolverhampton & Modul University Vienna*, UK/Austria **EUR 1,550,200**

EU Horizon Europe MSCA Staff Exchange project (guarantee funded by UKRI/EPSRC); UoW part GBP 218,036 (EUR 253,000). Topic: Tackling information overload. Project partners: Warsaw University of Technology (Poland, consortium lead), Wroclaw University of Science and Technology (Poland), Bar-Ilan University (Israel), University of Wolverhampton (UK), Modul University Vienna (Austria), Slovenian Press Agency (Slovenia), GESIS Leibniz Institute for the Social Sciences (Germany), Technion – Israel Institute of Technology (Israel); plus several associate partners in the US, UK, Singapore, Japan, China, Australia and The Netherlands. Co-authored the successful research proposal, specified the main work package on retrieval and recommendation to tackle information overload in academia. PI and work package leader, working on creating and evaluating AI-based search and recommendation technologies to avoid information overload. Above equal leading contribution as work package leader, researcher, supervisor and PI for the University of Wolverhampton (2023–2025) and Modul University Vienna (since 2025).

2023–2024 **RIGHT**, *University of Wolverhampton*, UK **GBP 612,621**

Internally funded project. Topic: deep learning for digital humanities. Equal leading contribution as PI and supervisor.

2023–2024 **LASER**, *University of Wolverhampton*, UK **GBP 39,000**

Responsible Digital Humanities Lab. Internally funded PhD project. Topic: Large language models in information retrieval. PI and single proposal author.

2016–2020 **QUARTZ**, *University of Bedfordshire*, Luton, UK **EUR 3,461,495**

EU H2020 Marie Skłodowska-Curie ITN European Training Network; own part GBP 473,387 (EUR 546,576). Topic: quantum-based information access and retrieval. Other project partners: University of Padua (Italy, consortium lead), The Open University (UK), Vrije Universiteit Brussel (Belgium), University of Copenhagen (Denmark), Brandenburg University of Technology (Germany), Linnæus University (Sweden). Co-authored the successful research proposal and specified a work package on interactive quantum-inspired retrieval models. PI and work package leader, working on multimodal retrieval models inspired by quantum mechanics and information foraging theory. Above equal leading contribution as work package leader, supervisor of early stage researchers, and PI.

2012–2013 **Rising Star**, *University of Bedfordshire*, Luton, UK **GBP 6,000**

Internal Research Investment Rising Star Funding. Topic: Query session processing for information seeking and searching. PI and working on information retrieval approaches. Single proposal author.

2010–2011 **QONTEXT**, *University of Glasgow*, Glasgow, UK

EU Marie Curie FP7-IRSES project. Topic: information retrieval and quantum theory. Other partners: University of Aberdeen (UK), University of Padua (Italy) and Open University (UK). Co-authored the successful research proposal. Worked on quantum-inspired retrieval models. Equal contribution.

2004–2006 **CLASSIX**, *University of Duisburg-Essen*, Duisburg, Germany

German Research Foundation (DFG) project. Topic: classification and intelligent search on information in XML. Other partner: University of Saarbrücken (Germany). PI and working on annotation-based retrieval models with structured documents. Main contributor (80%).

Knowledge Transfer and Industry Grants (as PI / Lead)

I led the successful bids and have been the Principal Investigator (PI) and lead academic.

2016–2019 **Drax Technology Ltd**, *University of Bedfordshire*, Luton, UK **GBP 134,752**

Innovate UK Knowledge Transfer Partnership with Drax Technology. Topic: Internet of Things, machine learning and intelligent fire alarms. Led and co-authored the successful bid (60% contribution). Lead academic and PI, supervising an MSc by Research student working on machine learning solutions to detect false fire alarms.

2012–2018 **NotActivelyLooking.com** **ca. GBP 20,000**

Private consultancy project (in collaboration with industry partners). Topic: design and implementation of a search engine for executives. Successfully negotiated with the client and acquired the project. PI and main software designer and developer. Main contributor (80%).

2016–2017 **Wardown Park Talking Objects**, *University of Bedfordshire*, Luton, UK **GBP 18,971**

With Wardown Park Museum, Luton (Luton Cultural Services Trust). Topic: combining augmented reality and conversational information retrieval. PI and supervision of an MSc-by-Research student working on augmented reality and information retrieval in a museum. Contribution around 40%.

2015 **EduVocation**, *University of Bedfordshire*, Luton, UK **GBP 5,000**
Teacher recruitment site and information system based on linked open data (UK Innovation Voucher). Led and co-authored the successful bid. PI and supervision of an MSc student working on search solutions for the website. The project led to the establishment of the successful MyNewTerm.com startup. Contribution around 40%.

2013–2015 **Luton Culture World War I project**, *University of Bedfordshire*, Luton, UK **GBP 16,000**
With Luton Museum (UK Lottery Fund). Digital Museum and Linked Data project on World War I. Led and co-authored the successful bid. PI, lead academic and supervisor of a software developer. Contribution around 40%.

2006 **KIIB**, *University of Duisburg-Essen*, Duisburg, Germany
EU-EUREGIO project in collaboration with d.velop AG, Gescher, Germany. Topic: business processes and email classification. Led and co-authored the successful bid. PI and main contributor (90%) to the resulting study.

Further Research Projects (Contributor)

In these projects, I contributed significantly to relevant work packages with my expertise.

2016–2017 **EDISON**, *University of Bedfordshire*, Luton, UK
EU H2020 Coordination and Support Action to accelerate the creation of the Data Science profession. UoB participated as Data Science Champion University, which I led and initiated. Presented and discussed our ideas to establish an undergraduate Data Science programme with international partners.

2008–2011 **RENAISSANCE**, *University of Glasgow*, Glasgow, UK
EPSRC project on retrieval models based on the quantum theory framework. Main task as postdoc: evaluation and further extension of an interactive retrieval framework inspired by quantum mechanics, formulation of the underlying theory.

2008 **LACOSTIR**, *University of Duisburg-Essen*, Duisburg, Germany
German Research Foundation (DFG) project on interactive retrieval. Main tasks: specification of a model for interactive retrieval, formulation of the underlying theory.

2006–2007 **DELOS**, *University of Duisburg-Essen*, Duisburg, Germany
EU-FP6 Network of Excellence on Digital Libraries. Main task: design and integration of annotation services into the agent-based Daffodil framework.

2004 **BRICKS**, *Fraunhofer IPSI*, Darmstadt, Germany
EU-FP6 integrated project on distributed resources for integrated cultural knowledge services. Main task: specification of annotation services.

2002–2003 **VIP-ADVISOR**, *Fraunhofer IPSI*, Darmstadt, Germany
EU-FP5 project (IST-2001-32440) on a virtual advisor for personal insurance and finance risk management. Main task: integration of case-based retrieval services for advisory dialogues.

2001–2003 **COLLATE**, *Fraunhofer IPSI*, Darmstadt, Germany
EU-FP5 project (IST-1999-20882) on the development of a collaboratory for annotation, indexing and retrieval of digitized historical archive material. Main task: design, implementation and evaluation of context-based retrieval methods based on user annotations, of retrieval Web services and of research prototypes.

2001–2003 **SCHOLNET**, *Fraunhofer IPSI*, Darmstadt, Germany
EU-FP5 project (IST-1999-20664) on a digital library infrastructure for supporting communication and collaboration among networked scholarly communities. Main task: design and implementation of search user interfaces.

Further Knowledge Transfer Projects (Contributor)

In these projects, I contributed significantly to relevant work packages with my expertise.

2020–2022 **SOLVD**, *University of Wolverhampton*, Wolverhampton, UK
EU European Regional Development Fund to support SMEs in the adoption of digital technologies. Main task: knowledge exchange (AI, automation, search technologies) with industry, and supervision.

2016 **CRASH**, *University of Bedfordshire*, Luton, UK
Comprehensible system for the classification of cyberstalking and harassment cases, in collaboration with Bedfordshire Police (Cyber Risk Assessment of Stalking and Harassment). Main task: supervision and design of a machine-learning-based framework against cyber stalking.

Teaching Experience

External Examiner and Programme Approval

since 2023 **External Examiner BSc Data Science**, *University of Sheffield*, Sheffield, UK

2020–2024 **External Examiner MSc Information Management and MSc Software Development**, *University of Strathclyde*, Glasgow, UK

2023 **MSc Computer Science Programme Approval Panel Member**, *University of Law*, London, UK

Programme Co-ordinator

since 2026 **MSc Data Science for Sustainability**, *Modul University Vienna*, Austria

since 2025 **BSc Applied Data Science**, *Modul University Vienna*, Austria

2021–2025 **MSc Computer Science and Data Science / MSc Computer Science and Artificial Intelligence**, *University of Wolverhampton*, UK

2016–2020 **BSc (Hons) Information and Data Systems**, *University of Bedfordshire*, Luton, UK

2014–2019 **BSc (Hons) Computing and Data Science**, *University of Bedfordshire*, Luton, UK

2011–2014 **BSc (Hons) Information Systems**, *University of Bedfordshire*, Luton, UK

2011–2014 **BSc (Hons) Business Information Systems**, *University of Bedfordshire*, Luton, UK

Module Co-ordinator and Lecturer

since 2026 **Information Retrieval**, *Modul University Vienna*, Austria
Lectures and lab sessions. Postgraduate module.

since 2026 **Machine Learning I**, *Modul University Vienna*, Austria
Lectures and lab sessions. Postgraduate module.

since 2025 **Foundations of Computer Science and Programming**, *Modul University Vienna*, Austria
Lectures and lab sessions. Undergraduate module.

since 2025 **Data Structures and Algorithms**, *Modul University Vienna*, Austria
Lectures and lab sessions. Undergraduate module.

2024 **Advanced Artificial Intelligence and Machine Learning**, *University of Wolverhampton*, UK
Lectures and lab sessions on machine learning, deep neural networks and generative AI. Postgraduate module.

2022–2024 **Big Data**, *University of Wolverhampton*, UK
Lectures and lab sessions on data analysis, information retrieval and Web search. 3rd year undergraduate module.

2023–2025 **Multivariate Statistics with Cybermetrics**, *University of Wolverhampton*, UK
Designing and delivering lectures on social media data analysis; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 3rd year undergraduate module.

since 2023 **Advanced Machine Learning**, *Bern University of Applied Sciences*, Biel/Bienne, Switzerland
Lectures and lab sessions on machine learning and information retrieval. 3rd year undergraduate module.

since 2023 **Interactive Data Intelligence**, *Bern University of Applied Sciences*, Biel/Bienne, Switzerland
Lectures and lab sessions on information retrieval. 3rd year undergraduate module.

since 2023 **Smart Data-Driven Applications**, *Bern University of Applied Sciences*, Biel/Bienne, Switzerland
Lectures and lab sessions on advanced information retrieval, retrieval-augmented generation and vector search. 3rd year undergraduate module.

2021–2024 **Data Management**, *University of Wolverhampton*, UK
Lectures and lab sessions on databases and SQL. Postgraduate module.

2022–2024 **Advanced Databases**, *University of Wolverhampton*, UK
Module co-ordinator; designing and delivering lectures on databases and advanced SQL; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 2nd year undergraduate module.

2023–2024 **Databases**, *University of Wolverhampton*, UK
Delivering lectures on databases and basic SQL; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 1st year undergraduate module.

2020–2024 **Database Systems and Security**, *University of Wolverhampton*, UK
Module co-ordinator; delivering lectures on data modelling, relational and NoSQL databases; responsible for organising laboratory sessions, student assignments and the general assessment of student progress

2019–2020 **Data Retrieval, Engineering and Visualisation**, *University of Bedfordshire*, Luton, UK
Module co-ordinator; delivering lectures on information and data retrieval, digital libraries and repositories; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 3rd year undergraduate module

2017–2020 **Intelligent Systems and Data Mining**, *University of Bedfordshire*, Luton, UK
Delivered lectures on information retrieval, data warehousing and NoSQL. Postgraduate module.

2017–2020 **Mathematics of Data**, *University of Bedfordshire*, Luton, UK
Delivered lectures on statistics for data analysis. 1st year undergraduate module.

2013–2020 **Advanced IT Strategies and Paradigms**, *University of Bedfordshire*, Luton, UK
Module co-ordinator; delivering lectures on information retrieval, XML and ontologies; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 3rd year undergraduate module.

2013–2020 **Principles of Programming and Data Structures**, *University of Bedfordshire*, Luton, UK
Module co-ordinator; designing and delivering lectures on Java programming; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; 1st year undergraduate module.

2011–2016 **(Advanced) Data Modelling and Management**, *University of Bedfordshire*, Luton, UK
Module co-ordinator; designing and delivering lectures on UML, databases and data modelling; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; postgraduate module. Responsible for delivery of lectures at partner college in Oman.

2014–2016 **Grid Services and Distributed Networking**, *University of Bedfordshire*, Luton, UK
Module co-ordinator; designing and delivering lectures on XML, Web Services, Hadoop, grid computing and cloud computing; responsible for organising laboratory sessions, student assignments and the general assessment of student progress; postgraduate module taught at partner college in Oman.

2010/11 **Distributed Information Management / Information Technology**, *University of Glasgow*, Glasgow, UK
Delivered lectures on data representation with XML. Postgraduate and 3rd year undergraduate module.

2010/11 **General Readings in Computer Science**, *University of Glasgow*, Glasgow, UK
Examined student assessments. Postgraduate module.

2006–2008 **Modelling Methods in Computer Science**, *University of Duisburg-Essen*, Duisburg, Germany
Delivered lectures and practicals on Petri Nets. Undergraduate module.

2008 **Information Retrieval**, *University of Duisburg-Essen*, Duisburg, Germany
Delivered lectures on information retrieval. Postgraduate module.

2008 **Information Mining**, *University of Duisburg-Essen*, Duisburg, Germany
Delivered lectures on machine learning and data mining. Postgraduate module.

2007 **Information Engineering**, *University of Duisburg-Essen*, Duisburg, Germany
Module co-ordinator; design and delivery of lectures and tutorials on methods, concepts and systems supporting the information life cycle (creation, management, dissemination and processing), which comprised authoring systems, metadata, databases, information and process modelling, digital preservation, access management, digital libraries and information seeking and retrieval; postgraduate module.

Research Student Supervision, Mentorship and Examination

Ongoing PhD Supervision

2025–2029 **Libo Ren: AI for Scholarly Access and Retrieval**, *Modul University Vienna*, Vienna, Austria
First supervisor

2026–2030 **Deniz Türkmen: Efficient Evaluation of Information Retrieval Systems in Real-World Settings through Living Labs and Generative Models**, *Technische Hochschule Cologne*, Cologne, Germany
External supervisor

2023–2026 **Junaid Muzaffar: A new transformer architecture to address permutation-invariant problems in reinforcement learning**, *University of Wolverhampton*, Wolverhampton, UK
External supervisor

2021–2027 **Oliver Cakebread-Andrews: Improving Sarcasm Detection Through Error Analysis and Building an English Language Corpus in Real Time**, *University of Wolverhampton*, Wolverhampton, UK
External supervisor

since 2023 **Patrick Steinert: Metaverse Recordings in Multimedia Information Retrieval**, *Fernuniversität Hagen*, Hagen, Germany
External supervisor

Completed PhD and MSc by Research Supervision

2023–2026 **Damith Premasiri: Legal information extraction**, *Lancaster University*, Lancaster, UK
External supervisor

2021–2025 **Kanishka Silva: Self-Attentive Generative Adversarial Network Based Authorship Attribution for Historical Texts**, *University of Wolverhampton*, Wolverhampton, UK
First supervisor of the successful PhD thesis of Dr Kanishka Silva, who became a postdoc at GESIS Leibniz Institute for the Social Sciences, Cologne, Germany.

2021–2025 **Khubaib Ahmed: Energy-efficient neural networks for speech enhancements**, *University of Wolverhampton*, Wolverhampton, UK
Director of studies. Khubaib Ahmed became research fellow at the University of Stirling, UK

2021–2025 **Mohsin Raza Naseem: Biologically plausible energy-efficient context-sensitive neural networks**, *University of Wolverhampton*, Wolverhampton, UK
Director of studies of the successful PhD thesis of Dr Mohsin Raza Naseem, who became a postdoc at the University of Stirling, UK.

2018–2023 **Investigation of Quantum-inspired Modelling in Interactive Search based on Information Foraging Theory**, *University of Bedfordshire*, Luton, UK
Second supervisor of the successful PhD thesis of Dr Amit Kumar Jaiswal. Dr Amit Kumar Jaiswal became a Postdoctoral Research Fellow at the University of Surrey, UK, and UCL, UK, and Assistant Professor at IIT Varanasi, India.

2017–2022 **A Theory-based Content-based Image Retrieval Approach for Capturing User Preferences during Query Formulation**, *University of Bedfordshire*, Luton, UK
Second supervisor of the successful PhD thesis of Dr Mohammed Mahmoud Artemi. Dr Mahmoud Artemi became Lecturer at the University of Bedfordshire, UK.

2012–2021 **AQUACOLD: Aggregated Query Understanding And Construction Over Linked Data**, *University of Bedfordshire*, Luton, UK
First supervisor of the successful PhD thesis of Dr Nick Collis (part time). Dr Collis became Senior Digital Education Support Analyst at Imperial College, London, UK.

2012–2020 **A Conceptual Framework to Support Cross-cultural User Experience Design for Web Search**, *University of Bedfordshire*, Luton, UK
First supervisor of the successful PhD thesis of Dr Karen Chessum (part time). Dr Chessum became Lecturer at the University of Bedfordshire, UK.

2012–2018 **A Computer Assisted Analysis of Literary Text: From Feature Analysis To Judgements of Literary Merit**, *University of Bedfordshire*, Luton, UK
Second supervisor of the successful PhD thesis of Dr Tess Crosbie. Dr Tess Crosbie became Senior Lecturer in Computer Science at the University of Bedfordshire, UK.

2013–2017 **A Methodology for Contextual Recommendation using Artificial Neural Networks**, *University of Bedfordshire*, Luton, UK
First supervisor of the successful PhD thesis of Dr Ghulam Mustafa. Dr Ghulam Mustafa became Assistant Professor at University of Central Punjab Lahore, Pakistan.

2011–2015 **A Probabilistic Approach for Cluster Based Interactive Information Retrieval**, *University of Bedfordshire*, Luton, UK
First supervisor of the successful PhD thesis of Dr Kamran Abbasi. Dr Abbasi became Associate Professor at the University of Sindh, Pakistan.

2011–2015 **A Framework for e-Government Success from the User's Perspective**, *University of Bedfordshire*, Luton, UK
Second supervisor of the successful PhD thesis of Dr Obaid Almalki. Dr Obaid Almalki works for the Saudi Arabian government.

2017–2020 **Use of machine learning to reduce false fire alarms**, *University of Bedfordshire*, Luton, UK
First supervisor of the MSc by Research project of MSc Usman Ali. Usman Ali became Full Stack Developer at Filament AI in London, UK.

2017–2018 **Developing and Evaluating a Cost-effective Solution for Augmented Reality Conversational Agents for Museums**, *University of Bedfordshire*, Luton, UK
First supervisor of the successful MSc by Research project of MSc Sebastian Grygiel. Sebastian Grygiel became Software Engineer in London, UK.

PhD Mentorship

2011–2012 **Balancing Exploration and Exploitation in Learning to Rank Online**, *University of Amsterdam*, Amsterdam, The Netherlands
In the context of the ECIR 2011 student mentoring program I was appointed the mentor of Dr Katja Hofmann's successful PhD thesis. She became Senior Principal Researcher at Microsoft Research, Cambridge, UK.

2010–2011 **Human Information Processing Based Information Retrieval**, *University of Glasgow*, Glasgow, UK
I have been assisting in supervising the successful PhD of Prof. Dr. Erik Graf about human information processing and information retrieval (supervisor: M. Lalmas). Dr. Erik Graf became Professor at the Bern University of Applied Sciences, Switzerland.

PhD and MSc by Research Examination

ongoing **National expert and external examiner PhD project "Mining Strategic Communication from Corporate Web Archives at Global Scale" by Hernán Picatto**, *TU Wien*, Vienna, Austria

2026 **External examiner PhD Thesis "An Interdisciplinary Study of Social Media Conflict in Brand Communities: Advanced Classification Models and Data Augmentation Frameworks for Marketing Data Insights and Analysis" by Oliver Warke**, *University of Glasgow*, Glasgow, UK

2025 **External examiner PhD Thesis "Efficient Parameter Adaptation Methods for Large Language Models: A Unifying Framework" by Wei Li**, *University of Birmingham*, Birmingham, UK

2025 **External examiner PhD Thesis "Identifying and Annotating Scientific Uncertainty in Scholarly Texts: Methods, Frameworks, and Applications" by Panggih Kusuma Ningrum**, *Université Marie & Louis Pasteur*, Besançon, France

2025 **Depth Exam examiner PhD candidate Naghmeh Farzi on LLM-as-a-judge**, *University of New Hampshire*, Durham, New Hampshire, USA

2025 **Internal examiner PhD Thesis "A Culturally-Aware Privacy Enhancing Technologies Framework for Improving the E-learning Systems Usage in Libyan Universities" by Marwa Alghodi**, *University of Wolverhampton*, UK

2023 **External examiner PhD Thesis "Logic Based Models for Information Retrieval in Social Media Context" by Firas Sabbah**, *University of Duisburg-Essen*, Duisburg, Germany

2022 **External examiner PhD Thesis "Semantic-aware Retrieval Standards based on Dirichlet Compound Model to Rank Notifications by Level of Urgency" by Mohammad Bahrani**, *Queen Mary, University of London*, UK

2022 **External examiner PhD Thesis "Understanding Novice Programmer Behavior on Introductory Courses – Learning Analytics Approach" by Erno Lokkila**, *University of Turku*, Finland

2022 **Internal examiner PhD Thesis "Modified mean and quantile regression models for citation analysis" by Marzieh Shahmandi**, *University of Wolverhampton*, UK

2021 **External examiner PhD Thesis "Semantic Classification and Evaluation" by Lukas Chaves Lima**, *University of Copenhagen*, Copenhagen, Denmark

2021 **External examiner PhD Thesis "Binary Classifier Inspired by Quantum Detection Theory" by Prayag Tiwari**, *University of Padua*, Padua, Italy

2020 **Internal examiner PhD Thesis "Energy Efficient Technique for Hadoop MapReduce Cluster Management" by Manal Alalawi**, *University of Bedfordshire*, Luton, UK

2019 **External examiner PhD Thesis "Language-Independent Methods for Identifying Cross-Lingual Similarity in Wikipedia" by Monica Paramita**, *University of Sheffield*, Sheffield, UK

2018 **External examiner MRes "User Engagement with Generous Interfaces for Digital Cultural Heritage" by Rob Speakman**, *Edge Hill University*, Ormskirk, UK

2016 **Internal examiner PhD Thesis "Using Social Media for Supporting Decision-Making in Managing Public Relations: The Case of Abu Dhabi Police" by Mansour Alkhyeli**, *University of Bedfordshire*, Luton, UK

2015 **External examiner PhD Thesis "Co-occurrence Rate Networks – An alternative theory for undirected graphical models" by Zhemin Zhu**, *University of Twente*, Enschede, The Netherlands

2015 **External examiner PhD Thesis "Semantic Models for Question Answering" by Piero Molino**, *Dipartimento di Informatica, Università Degli Studi Di Bari Aldo Moro*, Bari, Italy

2023–2025 **Internal reviewer for annual PhD progress (APR)**, *University of Wolverhampton*, UK
Internal reviewing process of University of Wolverhampton's PhD students.

MSc and BSc Project Supervision

since 2008, ongoing Supervised several (> 100) MSc and BSc projects in Austria, Germany and the UK on machine learning, information retrieval, natural language processing, data analysis, cybersecurity, high-throughput computing, personal digital assistants, business information systems, distributed systems and computer networks.

Keynotes, Invited Talks and Tutorials

2026 **Keynote on Information Overload in Academia**, *SESAME workshop co-located with JCDL 2026*, Dallas, Texas, USA

2026 **Keynote on Scholarly Information Access**, *SCOLIA Workshop at ECIR 2026*, Delft, The Netherlands

2026 **Invited Expert**, *Dagstuhl Seminar 26382, Frontiers of Technology Assisted Review Systems*, Schloß Dagstuhl – Leibniz-Zentrum für Informatik GmbH, Germany
Invitation-only seminar participation.

2025 **Keynote on Scholarly Information Access**, *Forum for Information Retrieval Evaluation (FIRE 2025), Indian Institute of Technology (BHU)*, Varanasi, India

2025 **Invited Expert**, *Dagstuhl Seminar 25381, Open Scholarly Information Systems: Status Quo, Challenges, Opportunities*, Schloß Dagstuhl – Leibniz-Zentrum für Informatik GmbH, Germany
Invitation-only seminar participation.

2025 **Tutorial on Information Retrieval Models and Evaluation**, *ELLIS European Summer School on Information Retrieval (ESSIR)*, Wolverhampton, UK

2024 **Keynote on Retrieval-Augmented Generation**, *Shaping The Future Of Our Society Through AI And Data Science Workshop, University of Derby*, Derby, UK

2022 **Tutorial on Information Retrieval Evaluation**, *BCS/ISKO Exploring Information Retrieval Lecture Series*, online

2021 **Practitioners' Evaluation Roundtable**, *BCS Search Solutions*, London, UK
Interactive tutorial on evaluation in academia and industry.

2014–2019 **Tutorial series on information retrieval models**, *Information Retrieval and Foraging Autumn School*, Schloß Dagstuhl and Katlenburg Castle, Germany

2015 **Keynote on Polyrepresentation in Complex (Book) Search Tasks**, *6th International Conference of the CLEF Association (CLEF 2015), Social Book Search Lab*, Toulouse, France

2012 **Tutorial on quantum probabilities and quantum information access and retrieval**, *Information Retrieval Autumn School*, Schloß Dagstuhl, Germany

ongoing **Invited talks**, *Several invited talks at university and industry seminars*
Queen Mary, University of London (2001, on automatic categorisation of Web documents, 2004, on annotation-based information retrieval, 2010, on probabilistic logics, 2020, on quantum-inspired retrieval and polyrepresentation); University of Padua (2006, on annotation-based information retrieval); University of Duisburg-Essen (2010, on quantum-inspired information retrieval); UCL London (2012, on quantum-inspired retrieval); Maxeler Technologies Ltd., London (2015, on Twitter Sentiment Analysis with FPGAs); University of Cologne (2015, on information access in digital humanities); University of Nottingham (2016, on quantum-inspired information retrieval); Lumi Ltd., London (2016, on Context, Interaction, Logics and Probability for Complex Search Tasks); Signal Media Ltd., London (2017, on quantum-inspired information retrieval); University of Regensburg (2017, on interactive information retrieval); University of Glasgow (2020, on multimodal information retrieval); Refinitiv (now LSEG Data & Analytics), London (2021, on quantum-inspired information retrieval); University of Southampton (2022, on interactive information access); University of New Hampshire (2022, on quantum-inspired retrieval models and deep learning); University of Exeter (2024, on AI in multimodal interactive information retrieval models); Modul University Vienna (2024, on interactive information retrieval); WU – Vienna University of Economics and Business (2025, on interactive information retrieval and scholarly information access); University of Southampton, University of Bedfordshire, University of Sheffield, University of Glasgow, University of Wolverhampton, UK (2026, series of invited talks on academic information overload and scholarly information access); Paris Lodron University of Salzburg, Austria (2026, lecture on interactive information access and generative AI); National University of Singapore and Nanyang Technological University, Singapore (2026, academic information overload and scholarly information access).

Leadership, Management and Steering Roles

since 2025 **Head of School of Applied Data Science**, *Modul University*, Vienna, Austria

since 2025 **Chair (Obmann)**, *Vienna Network for Research and Innovation*, Vienna, Austria

since 2024 **Chair of the BCS Information Retrieval Specialist Group (BCS-IRSG)**, *The Chartered Institute for IT (BCS)*, UK

2014–2024 **Treasurer, BCS-IRSG**, *The Chartered Institute for IT (BCS)*, UK

since 2011 **Steering committee member, BCS-IRSG**, *The Chartered Institute for IT (BCS)*, UK

2023–2025 **Head of Data Science and AI Research**, *University of Wolverhampton*, UK
Head and founder of Wolverhampton's Data Science and AI Research group DAIREL.

2023–2025 **Deputy Director of Digital Innovation and Services Centre (DISC)**, *University of Wolverhampton*, UK

since 2024 **Steering committee member, ACM CIKM Conference**
ACM Conference on Information and Knowledge Management, Core-A ranked

since 2024 **Steering committee member, ACM SIGIR ICTIR Conference**
ACM International Conference on Theory in Information Retrieval.

2008–2018 **Steering committee member, German Computing Society Information Retrieval Special Interest Group**
Elected member of the steering committee of the IR Special Interest Group of the German Computing Society (Fachgruppe Information Retrieval der Gesellschaft für Informatik).

2018–2020 **Information and Data Science Subject Cluster Lead**, *University of Bedfordshire*, Luton, UK
Coordinating the Information and Data Science research and teaching activities.

2017–2018 **BCS Accreditation**, *University of Bedfordshire*, Luton, UK
Coordinated the successful BCS accreditation application of undergraduate and postgraduate computer science programmes taught at the University of Bedfordshire.

2016 **VC's Research & Enterprise Sub Group**, *University of Bedfordshire*, Luton, UK
Member of the Vice Chancellor's sub group to shape the university's research and enterprise strategy.

2013–2014 **Data Science Programme Approval**, *University of Bedfordshire*, Luton, UK
Responsible for the design and specification of the successfully approved undergraduate BSc (Hons) Data Science programme at the University of Bedfordshire, including the design and specification of a module on Data Engineering and Retrieval.

2013–2015 **Staff Representative**, *University of Bedfordshire, Institute for Research in Applicable Computing (IRAC)*, Luton, UK
Elected staff representative in the IRAC steering committee.

2013–2015 **Deputy Head**, *IRAC Centre for Research in Distributed Technologies (CREDIT), University of Bedfordshire*, Luton, UK

Conference Chairing and Event Organisation

I have been organising several prestigious international conferences, workshops and schools, as general chair (notably ACM CIKM 2023) or main organiser, as well as chair or senior programme committee member.

General Chair and Organiser

2025 **ELLIS ESSIR 2025 – 16th European Summer School on Information Retrieval**, *July 7–11, 2025, Wolverhampton*, UK
General chair and in charge of the scientific programme, including Information Retrieval and AI topics.

2022–2023 **CIKM 2023 – ACM International Conference on Information and Knowledge Management**, *October 21–25, 2023, Birmingham*, UK
Leading General Chair. Core-A-ranked conference with 750 registered delegates, approx. 650 accepted papers from ca. 2500 submissions, and world-leading keynote speakers.

2015–2026 **BIR/BIRNDL/SCOLIA – International Workshop on Bibliometric-enhanced Information Retrieval; The Joint Workshop on Bibliometric-enhanced Information Retrieval and Natural Language Processing for Digital Libraries; later renamed to SCOLIA – Scholarly Information Access**
Co-organiser of the workshop series held at ECIR, JCDL and SIGIR conferences 2015–2026. Guest editor of corresponding special issues at IJDL and Scientometrics.

2016–2026 **BCS Search Solutions**, *London*, UK
Co-organiser of the annual BCS Search Solutions Conference on Innovations in Search & Information Retrieval, addressed to industry practitioners and academics.

2020–2021 **BIRDS – International Workshop on Bridging the Gap Between Information Science, Information Retrieval and Data Science**
Co-organiser of the interdisciplinary Information Science, Information Retrieval and Data Science workshop at SIGIR 2020 and CHIIR 2021.

2017–2024 **FDIA – Future Directions in Information Access**
Co-organiser of the symposium for PhD students in Information Retrieval. 2017 (Barcelona, Spain), 2018 (Tianjin, China), 2019 (Milan, Italy), 2022 (Lisbon, Portugal), 2023 (Vienna, Austria), 2024 (Amsterdam, The Netherlands).

2019 **TALMIRI – TALent Meets Information Retrieval Industry**, *Luton*, UK
Organised a 1-day symposium to bring together academia and industry.

2014–2019 **Information Retrieval and Foraging Autumn School**, Germany
Co-organiser of annual one week schools on selected information foraging and retrieval topics (2014, 2016–2019 at Schloß Dagstuhl, Germany, 2015 at Castle Katlenburg, Germany)

2006 **Heterogeneous Track at the Initiative for the Evaluation of XML Retrieval**
Co-organiser of the INEX 2006 Heterogeneous Track.

Programme and Other Chairing Roles

2027 **European Conference on Information Retrieval (ECIR 2027)**, *Southampton*, UK
Short Paper Programme Chair.

2026 **ACM/IEEE Joint Conference on Digital Libraries (JCDL 2026)**, *Dallas*, Texas, USA
Panel Chair.

2025 **ACM/IEEE Joint Conference on Digital Libraries (JCDL 2025)**, *online*
Programme Chair.

2025 **ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR 2025)**, *Melbourne*, Australia
Best paper committee chair.

2022 **ACM/IEEE Joint Conference on Digital Libraries (JCDL 2022)**, *Cologne*, Germany
Senior PC Member and Publicity Chair.

2019 **European Conference on Information Retrieval (ECIR 2019)**, *Cologne*, Germany
Publicity Chair.

2018 **European Conference on Information Retrieval (ECIR 2018)**, *Grenoble*, France
Publicity Chair.

2018 **ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR 2018)**, *Tianjin*, China
Publicity Chair.

2016 **ACM/IEEE Joint Conference on Digital Libraries (JCDL 2016)**, *Newark*, NJ, USA
Publicity Chair.

2016 **International Conference on Theory and Practice of Digital Libraries (TPDL 2016)**, *Hannover*, Germany
Tutorial chair.

2015 **European Conference on Information Retrieval (ECIR 2015)**, *Vienna*, Austria
Student mentor.

2014 **DL2014: ACM/IEEE Joint Conference on Digital Libraries (JCDL 2014) / International Conference on Theory and Practice of Digital Libraries (TPDL 2014)**, *London*, UK
Senior PC member and chair of the best student paper committee. In 2014 JCDL and TPDL were held together as the International Digital Libraries Conference (DL2014).

2015–2018 **IR workshops at Lernen–Wissen–Adaptivität (LWA-WIR)**

2009–2012 Programme chair of the IR workshops held by the IR Special Interest Group of the German Computing Society (GI) in 2009 (Darmstadt), 2010 (Kassel), 2011 (Magdeburg), 2012 (Dortmund), 2015 (Trier), 2016 (Potsdam).

2013 **International Conference on Theory and Practice of Digital Libraries (TPDL 2013)**, *Valletta*, Malta
Panel chair.

2011 **International Symposium on Quantum Interaction (QI 2011)**, *Aberdeen*, UK
Proceedings chair.

2010 **European Conference on Research and Advanced Technology for Digital Libraries (ECDL 2010)**, *Glasgow*, UK
Local chair.

Senior Programme Committee Membership

ongoing **Senior PC Member**
Senior PC Member (Meta Reviewer) for several world-leading conferences, such as ACM SIGIR, ACM CIKM, ACM SIGIR-AP, ACM JCDL, ACM WSDM, TPDL

Programme Committee Membership and Reviewing

ongoing Member of the programme committee and reviewer at the following conferences and workshops: ACM SIGIR, ACM CIKM, ACM/IEEE JCDL, ACM ICTIR, ACM CHIIR, ACM WSDM, ACL, WWW, ECIR, ECDL/TPDL, IEEE Big Data, IIiX, Russian IR Summer School, Quantum Interaction, TIR, KDIR, LWA, LSHC, AIRS, HPCS.

ongoing Reviewer for several journals such as ACM Transactions on Information Systems (TOIS), Elsevier Information Systems, Elsevier Information Processing & Management (IP&M), The Journal of the American Societ Association for Information Science and Technology (JASIST), The Information Retrieval Journal (Springer), The Computer Journal, PLOS ONE, Theoretical Computer Science (Elsevier), Scientometrics (Springer), ACM Transactions on Computer-Human Interaction (TOCHI), Information Retrieval Research Journal (IRRJ) and The International Journal on Digital Libraries (Springer).

Further Academic Service and Engagement.

Grant and Proposal Evaluation

since 2019 Evaluator for Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)

since 2022 Evaluator for UKRI Engineering and Physical Sciences Research Council (EPSRC)

2024 Evaluator for NGI Search (Next Generation Internet Discovery and Search, EU Horizon Europe RIA Project)

Academic Promotion Reviews

2026 **External reviewer for the promotion of Dr Mat Kelly from Assistant to Associate Professor**, *Drexel University*, Philadelphia, USA

2026 **External reviewer for the promotion of Dr Ayesha Afzal from Assistant Professor to Tenured Associate Professor**, *Air University*, Islamabad, Pakistan

Editorial Boards and Guest Editorships

since 2001 **Senior Managing Editor and Editorial Board Member, International Journal on Digital Libraries (IJDL, Springer; 5-year Impact Factor: 1.8)**

since 2009 **Editorial Board Member, Datenbank-Spektrum Journal (Springer)**

2024 **Guest Editor IJDL Special Issue on Scholarly AI and Large Language Models**

2018–2020 **Guest Editor of several Scientometrics Special Issues on Bibliometrics and Information Retrieval**

2017 **Guest Editor IJDL Focussed Issue on Bibliometrics and Information Retrieval**

2011 **Guest Editor IJDL Focussed Issue on ECDL 2010**

Equity, Diversity, Inclusion and Community Engagement

since 2024 **Conference Ambassador**, *Birmingham & West Midlands Convention Bureau*, West Midlands Growth Company, UK
Contributing to attracting conferences in Data Science and AI to Birmingham and the West Midlands.

2022–2024 **CARA Mentorship Scheme**, *Council for At-Risk Academics (CARA)*, UK
As part of the scheme, I provided mentorship for a researcher from Ukraine.

2021 **Athena SWAN**, *School of Engineering, Computing and Mathematical Sciences*, University of Wolverhampton, UK
Part of the Athena SWAN Task Force to represent women in Science, Technology, Engineering, Mathematics and Medicine (STEMM). Provided a data analysis on gender balance in admissions in the UK and the university, for the school's Bronze Athena SWAN award application.

Professional Memberships and Fellowships

since 2014 **BCS Fellow (FBCS)**, *The Chartered Institute for IT (BCS)*, UK
`https://www.bcs.org/membership-and-registrations/become-a-member/bcs-fellowship/`

since 2016 **ACM Member (ACM SIGIR)**, *Association for Computing Machinery*, USA

since 2002 **GI Member**, *Gesellschaft für Informatik*, Germany

since 2012 **Fellow of the Higher Education Academy / Advance HE (FHEA)**, UK
`https://www.advance-he.ac.uk/fellowship/fellowship`

since 2016 **Member of the Engineering Professors' Council (MEPC)**, UK

Technical Skills

Hands-on with Python, LLM/RAG tooling, SQL, machine learning and search engine technologies, and IR data and evaluation pipelines.

Full List of Publications

The list begins with peer-reviewed conference publications followed by journal publications, preprints, proceedings and book sections, and theses, each in reverse chronological order. Citation information can be found on my Google Scholar profile.

Peer-Reviewed and Conference Publications

[1] Ingo Frommholz, Christin Katharina Kreutz, Philipp Mayr, and Guillaume Cabanac. The Second International Workshop on Scholarly Information Access (SCOLIA 2026). In Ricardo Campos, Adam Jatowt, Yanyan Lan, Mohammad Aliannejadi, Christine Bauer, Sean MacAvaney, Avishek Anand, Zhaochun Ren, Suzan Verberne, Nan Bai, and Masoud Mansoury, editors, *Proceedings of the European Conference on Information Retrieval (ECIR 2026)*, pages 204–212, Cham, 2026. Springer Nature Switzerland.

[2] Stephen Toriola, Anirban Chakraborty, and Ingo Frommholz. ImmiGo - An Adaptive Multilingual Chatbot for Immigration Assistance. In *Proceedings ACM SIGIR CHIIR 2026*, Seattle, United States, 2026. ACM Press.

[3] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Multimodal RAG Enhanced Visual Description. In *Proceedings of the ACM Conference on Information and Knowledge Management (CIKM 2025)*, Seoul, Korea, August 2025. ACM. arXiv:2508.09170 [cs].

[4] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, Suzan Verberne, and Christin Katharina Kreutz. The First Workshop on Scholarly Information Access (SCOLIA). In Claudia Hauff, Craig Macdonald, Dietmar Jannach, Gabriella Kazai, Franco Maria Nardini, Fabio Pinelli, Fabrizio Silvestri, and Nicola Tonellotto, editors, *Proceedings of the European Conference on Information Retrieval (ECIR 2025)*, pages 326–331, Cham, 2025. Springer Nature Switzerland.

[5] Oliver Cakebread-Andrews, Le An Ha, Ingo Frommholz, and Burcu Can. Error Analysis of NLP Models and Non-Native Speakers of English Identifying Sarcasm in Reddit Comments. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 6247–6256, Torino, Italia, May 2024. ELRA and ICCL.

[6] Patrick Steinert, Stefan Wagenpfeil, Ingo Frommholz, and Matthias L. Hemmje. Integration of Metaverse Recordings in Multimedia Information Retrieval. In *Proceedings of the 2024 13th International Conference on Software and Computer Applications*, ICSCA '24, pages 137–145, New York, NY, USA, May 2024. Association for Computing Machinery.

[7] Kanishka Silva, Ingo Frommholz, Burcu Can, Fred Blain, Raheem Sarwar, and Laura Ugolini. Forged-GAN-BERT: Authorship Attribution for LLM-Generated Forged Novels. In Neele Falk, Sara Papi, and Mike Zhang, editors, *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2024): Student Research Workshop*, pages 325–337, St. Julian's, Malta, March 2024. Association for Computational Linguistics.

[8] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, and Suzan Verberne. Bibliometric-Enhanced Information Retrieval: 14th International BIR Workshop (BIR 2024). In Nazli Goharian, Nicola Tonellotto, Yulan He, Aldo Lipani, Graham McDonald, Craig Macdonald, and Iadh Ounis, editors, *Proceedings of the European Conference on Information Retrieval (ECIR 2024)*, pages 442–446, Cham, 2024. Springer Nature Switzerland.

[9] Julius Odede and Ingo Frommholz. JayBot – Aiding University Students and Admission with an LLM-based Chatbot. In *Proceedings ACM SIGIR Conference on Human Information Interaction And Retrieval (CHIIR 2024)*, pages 391–395. ACM, 2024.

[10] Patrick Steinert, Stefan Wagenpfeil, Ingo Frommholz, and Matthias Hemmje. 256 Metaverse Recording Dataset. In *ACM Multimedia 2024*, 2024.

[11] Hamed Alhoori, Edward A. Fox, Ingo Frommholz, Haiming Liu, Corinna Coupette, Bastian A. Rieck, Tirthankar Ghosal, and Jian Wu. Who can Submit an Excellent Review for this Manuscript in the Next 30 Days? - Peer Reviewing in the Age of Overload. In *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, pages 319–320, Santa Fe, NM, USA, June 2023. IEEE.

[12] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, Suzan Verberne, and Jordan Brennan. Preface to the 13th Workshop on Bibliometric-enhanced Information Retrieval at ECIR 2023. In Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, Suzan Verberne, and Jordan Brennan, editors, *Proceedings of the 13th International Workshop on Bibliometric-enhanced Information Retrieval*, volume 3617 of *CEUR Workshop Proceedings*, pages 1–3, Dublin, Ireland, April 2023. CEUR.

[13] Karen Chessum, Haiming Liu, and Ingo Frommholz. An Extended Study of Search User Interface Design Focused on Hofstede's Cultural Dimensions. In Andreas Holzinger, Hugo Plácido Da Silva, Jean Vanderdonckt, and Larry Constantine, editors, *Computer-Human Interaction Research and Applications*, volume 1882, pages 130–152, Cham, 2023. Springer Nature Switzerland. Series Title: Communications in Computer and Information Science.

[14] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, and Suzan Verberne. Bibliometric-enhanced information retrieval: 13th international BIR workshop (BIR 2023). In Jaap Kamps, Lorraine Goeuriot, Fabio Crestani, Maria Maistro, Hideo Joho, Brian Davis, Cathal Gurrin, Udo Kruschwitz, and Annalina Caputo, editors, *Proceedings of the European Conference on Information Retrieval (ECIR 2023)*, pages 392–397, Cham, 2023. Springer Nature Switzerland.

[15] Kanishka Silva and Ingo Frommholz. What if ChatGPT wrote the Abstract? - Explainable Multi-Authorship Attribution with a Data Augmentation Strategy. In *Proceedings of the IACT - The 1st International Workshop on Implicit Author Characterization from Texts for Search and Retrieval (IACT@SIGIR 2023)*, pages 38–48, Taipei, Taiwan, 2023. CEUR-WS.org.

[16] Patrick Steinert, Stefan Wagenpfeil, Ingo Frommholz, and Matthias L. Hemmje. Towards the Integration of Metaverse and Multimedia Information Retrieval. In *2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE)*, pages 581–586. IEEE, 2023.

[17] Karen Chessum, Haiming Liu, and Ingo Frommholz. A Study of Search User Interface Design Based on Hofstede's Six Cultural Dimensions. In *Proceedings 6th International Conference on Computer-Human Interaction Research and Applications (CHIRA 2022)*, pages 145–154, Valetta, Malta, October 2022. SciTePress.

[18] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, and Suzan Verberne. Bibliometric-enhanced Information Retrieval: 12th International BIR Workshop (BIR 2022). In *Proceedings European Conference on Information Retrieval (ECIR 2022)*. Springer, 2022.

[19] Nicholas Collis and Ingo Frommholz. AQUACOLD: A Novel Crowdsourced Linked Data Question Answering System. In Ingo Frommholz, Haiming Liu, and Massimo Melucci, editors, *Proceedings Second Workshop on Bridging the Gap between Information Science, Information Retrieval and Data Science (BIRDS 2021)*, pages 11–25. CEUR-WS.org, 2021.

[20] Nicholas Collis and Ingo Frommholz. Crowdsourced Linked Data Question Answering with AQUACOLD. In *Procedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL 2021)*. ACM, 2021.

[21] Ingo Frommholz, Haiming Liu, and Massimo Melucci. BIRDS 2021: Bridging the Gap between Information Science, Information Retrieval and Data Science. In Falk Scholer, Paul Thomas, David Elsweiler, Hideo Joho, Noriko Kando, and Catherine Smith, editors, *CHIIR '21: Proceedings of the 2021 Conference on Human Information Interaction and Retrieval*, pages 349–350, Canberra, Australia, 2021. ACM.

[22] Ingo Frommholz, Philipp Mayr, Guillaume Cabanac, and Suzan Verberne. Bibliometric-enhanced Information Retrieval: 11th International BIR Workshop. In *Proceedings of the European Conference on Information Retrieval (ECIR 2021)*. Springer, 2021.

[23] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Semantic Hilbert Space for Interactive Image Retrieval. In *ICTIR 2021 - Proceedings of the 2021 ACM SIGIR International Conference on Theory of Information Retrieval*, pages 307–315, 2021.

[24] Guillaume Cabanac, Ingo Frommholz, and Philipp Mayr. Bibliometric-Enhanced Information Retrieval 10th Anniversary Workshop Edition. In *Proceedings European Conference on Information Retrieval (ECIR 2020)*, pages 641–647, Lisbon, Portugal, 2020.

[25] Ingo Frommholz, Haiming Liu, and Massimo Melucci. BIRDS-Bridging the Gap between Information Science, Information Retrieval and Data Science. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2455–2458. ACM, 2020.

[26] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Reinforcement Learning-driven Information Seeking: A Quantum Probabilistic Approach. In Ingo Frommholz, Haiming Liu, and Massimo Melucci, editors, *Proceedings First Workshop on Bridging the Gap between Information Science, Information Retrieval and Data Science (BIRDS 2020)*, pages 16–29. CEUR-WS.org, 2020.

[27] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Utilising Information Foraging Theory for User Interaction with Image Query Auto-Completion. In *Proceedings European Conference on Information Retrieval (ECIR 2020)*. Springer, 2020.

[28] Guillaume Cabanac, Ingo Frommholz, and Philipp Mayr. Bibliometric-Enhanced Information Retrieval: 8th International BIR Workshop. In *Proceedings European Conference on Information Retrieval (ECIR 2019)*, pages 394–399, Cologne, Germany, 2019. Springer.

[29] Ingo Frommholz, Haiming Liu, and Yashar Moshfeghi. The 9th symposium on future directions in information access. In Ingo Frommholz, Haiming Liu, and Yashar Moshfeghi, editors, *Proceedings of the 9th PhD Symposium on Future Directions in Information Access co-located with 12th European Summer School in Information Retrieval (ESSIR 2019)*, pages 1–3, Milan, Italy, 2019. CEUR-WS.org.

[30] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Effects of Foraging in Personalized Content-based Image Recommendation. In *Proceedings of the 2nd International Workshop on ExplainAble Recommendation and Search (EARS 2019) at SIGIR 2019*, 2019. arXiv: 1907.00483.

[31] Amit Kumar Jaiswal, Haiming Liu, and Ingo Frommholz. Information Foraging for Enhancing Implicit Feedback in Content-based Image Recommendation. In *Proceedings of the 11th Forum for Information Retrieval Evaluation (FIRE 2019)*, pages 65–69, Kolkata, India, 2019. ACM.

[32] Amit Kumar Jaiswal, Guilherme Holdack, Ingo Frommholz, and Haiming Liu. Quantum-like Generalization of Complex Word Embedding: a lightweight approach for textual classification. In *Proceedings Lernen Wissen Daten Analysen (LWDA 2018)*, 2018.

[33] Waleed Alabbas, Haider Al-Khateeb, Ali Mansour, Gregory Epiphaniou, and Ingo Frommholz. Classification of Colloquial Arabic Tweets in real-time to detect high-risk floods. In *Proceedings International Conference on Social Media, Wearable and Web Analytics (Social Media 2017)*. IEEE, 2017.

[34] Nicholas Collis and Ingo Frommholz. Aquacold - a crowdsourced query understanding and query construction tool for the linked data web. In Philipp Mayr, Douglas Tudhope, Koraljka Golub, Christian Wartena, and Ernesto William De Luca, editors, *Proc. of the 17th European Networked Knowledge Organization Systems Workshop (NKOS 2017) at TPDL 2017*, pages 74–86, Thessaloniki, Greece, 2017. CEUR-WS.org.

[35] Guillaume Cabanac, Muthu Kumar Chandrasekaran, Ingo Frommholz, Kokil Jaidka, Min-Yen Kan, Philipp Mayr, and Dietmar Wolfram. Joint workshop on bibliometric-enhanced information retrieval and natural language processing for digital libraries (BIRNDL 2016). In *2016 IEEE/ACM Joint Conference on Digital Libraries (JCDL)*, pages 299–300, June 2016.

[36] Faris Abomelha, Ingo Frommholz, and Paul Sant. A Study of Challenges and Obstacles in Biomedical Data Search. In *Proceedings Medical Information Retrieval (MedIR) Workshop at SIGIR 2016*, Pisa, Italy, 2016.

[37] P. Mayr, I. Frommholz, and G. Cabanac. Bibliometric-enhanced information retrieval: 3rd international BIR workshop. In *Proceedings of the European Conference on Information Retrieval (ECIR 2016)*, volume 9626, pages 865–868, 2016.

[38] Philipp Mayr, Ingo Frommholz, and Guillaume Cabanac. Editorial for the 3rd Bibliometric-Enhanced Information Retrieval Workshop at ECIR 2016. In *Proceedings BIR 2016 Workshop on Bibliometric-enhanced Information Retrieval*, pages 1–4. CEUR-WS.org, 2016.

[39] Ghulam Mustafa and Ingo Frommholz. Comparing contextual and non-contextual features in ANNs for movie rating prediction. In Ralf Krestel, Davide Mottin, and Emmanuel Müller, editors, *Proceedings of the Conference "Lernen, Wissen, Daten, Analysen", Workshop Information Retrieval*, Potsdam, Germany, 2016. CEUR-WS.org.

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